filtering the screening dataset a pisum

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Setup

Packages required

```
suppressPackageStartupMessages(library("tidyverse"))
suppressPackageStartupMessages(library("lme4"))
suppressPackageStartupMessages(library("Hmisc"))
suppressPackageStartupMessages(library("corrplot"))
suppressPackageStartupMessages(library("ggrepel"))
suppressPackageStartupMessages(library("patchwork"))
```

“

Import helper functions

```
source("../../broad_heritability_function.R")
source("../../create_correlation_plot.R")
```

```
Loading required package: glue
```

Broad sense heritability calculations

Raw dataset

```
data250pisum_00 <- read.csv("../../../../../../01.data/2023_screen_specifics_251genotypes/01
                            stringsAsFactors = FALSE,
                            sep = ",") %>%
  mutate(ivip = as.factor(ivip)) %>%
  mutate(session = as.factor(session))

res <- calculate_broad_heritability(data = data250pisum_00)
```

> Tip
>
> Conclusions:
> - Heritability is equal to 0.175.
> - There are 249 genotypes in total.
> - There are 3014 measurements in total.

we can't use our dataset like this, we need to clean it first.

Missing data removed

First we remove the missing data

```
data250pisum_01 <- data250pisum_00 %>% filter(!is.na(larvae_total_number))
data250pisum_01$ivip <- droplevels(data250pisum_01$ivip)

res2 <- calculate_broad_heritability(data = data250pisum_01)
```

> Tip
>
> Conclusions:
> - Heritability is equal to 0.175.
> - There are 247 genotypes in total.
> - There are 2637 measurements in total.

```
create_correlation_plot(data = data250pisum_01,
                        group_cols = c("ivip", "session"),
                        value_col = "larvae_total_number")
```

Remove observations with no adult alive

We chose to remove all data from plants on which no adult aphid was found. This decision was made to get rid of larvae installation bias and only assess the reproduction ability of the aphid.

When the adult aphid dies, the number of larvae will drop but only because the adult did not stay much time on the plant and not because the plant was not suitable for the reproduction.

```
data250pisum_02<- data250pisum_01 %>% filter(adult_alive != "0")
data250pisum_02$ivip <- droplevels(data250pisum_02$ivip)

res3 <- calculate_broad_heritability(data = data250pisum_02)
```

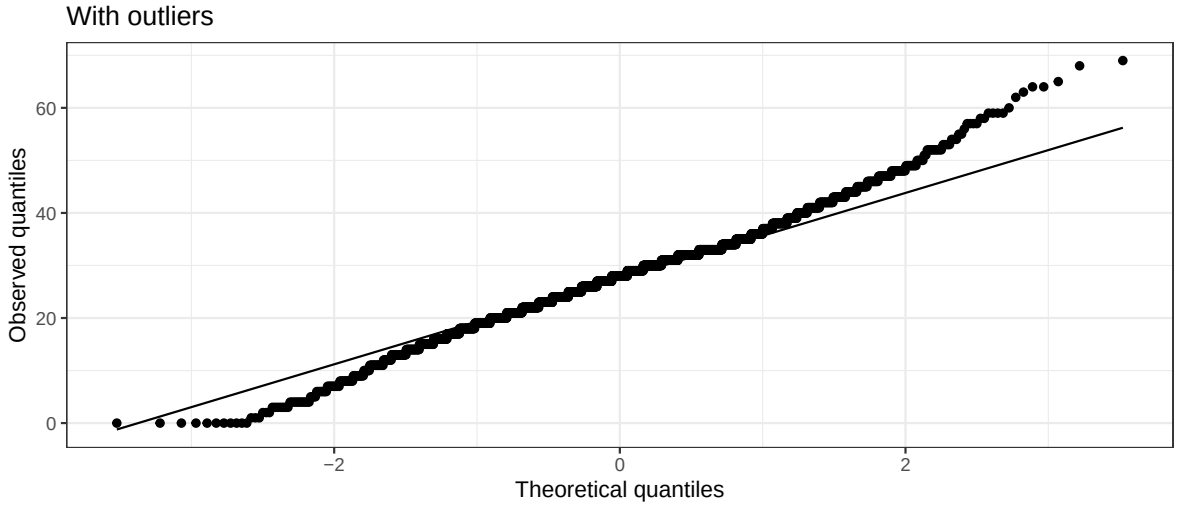
> Tip
>
> Conclusions:
> - Heritability is equal to 0.371.
> - There are 246 genotypes in total.
> - There are 2333 measurements in total.

This filtering led to the loss of 1 genotype but increased broad sense heritability up to 0.37

```
# Call the function to generate and display the correlation plot
create_correlation_plot(data250pisum_02,
                        group_cols = c("ivip", "session"),
                        value_col = "larvae_total_number")
```

Distribution normality check.

```r
p_with <- data250pisum_02 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("With outliers") +
  theme_bw()
p_with
```



Filter outliers

Our data are still not normally distributed. We need to apply a new filter. To remove outliers we choose to use the interquartile range method (IQR) removing data that are considered too high or too low. Both case can be explained biologically, low larvae count may be explain by an adult that was hurt during the installation and a larvae count too high may be explained by 2 adult beeing installed on the plant instead of only one.

```r
IQR_value <- IQR(data250pisum_02$larvae_total_number)
lower_bound <- quantile(data250pisum_02$larvae_total_number, 0.25) - 1.5 * IQR_value
upper_bound <- quantile(data250pisum_02$larvae_total_number, 0.75) + 1.5 * IQR_value

# Filter out outliers
data250pisum_03 <- data250pisum_02[data250pisum_02$larvae_total_number >= lower_bound & data2

# Identify the rows that were removed (outliers)
```

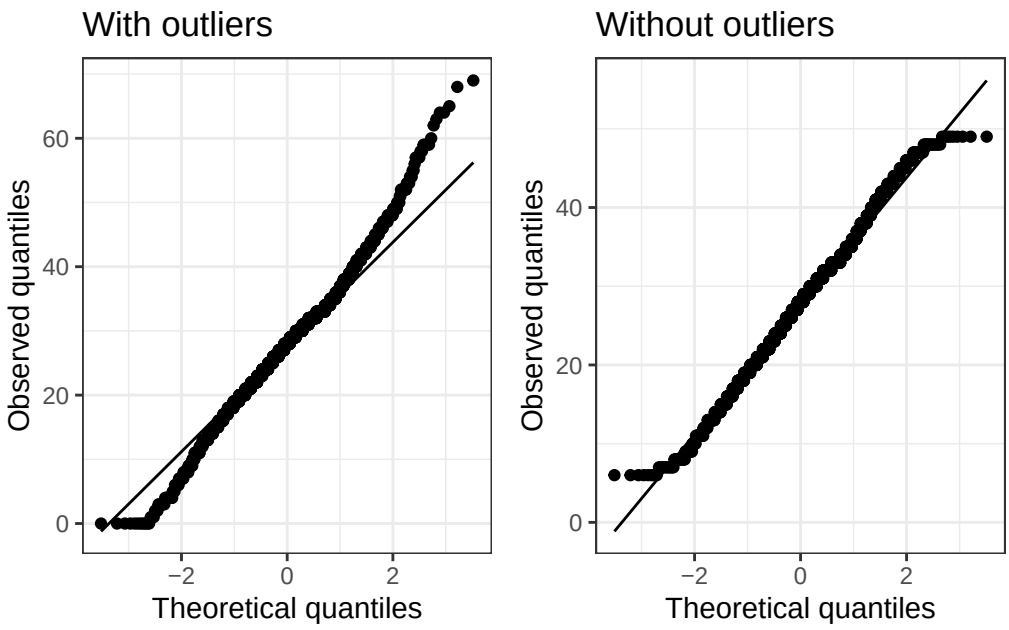
```r
removed_outliers <- data250pisum_02[data250pisum_02$larvae_total_number < lower_bound | data2

res4 <- calculate_broad_heritability(data = data250pisum_03)

#| fig-width: 8

# Without outliers
p_without <- data250pisum_03 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("Without outliers") +
  theme_bw()


p_with + p_without
```



Tip

Conclusions:
- Heritability is equal to 0.411.
- There are 246 genotypes in total.
- There are 2250 measurements in total.

Both the normality of the data and the heritability benefits from this filter.

Final filtering: keep genotypes with at least 4/5/6 biological repetitions

```
data250pisum_04_4 <- data250pisum_03 %>%
  group_by(ivip) %>%           # Group by genotype
  filter(n() >= 4) %>%             # Keep only groups with 4 or more repetitions
  ungroup()
data250pisum_04_4$ivip <- droplevels(data250pisum_04_4$ivip)
res5 <- calculate_broad_heritability(data = data250pisum_04_4)
```

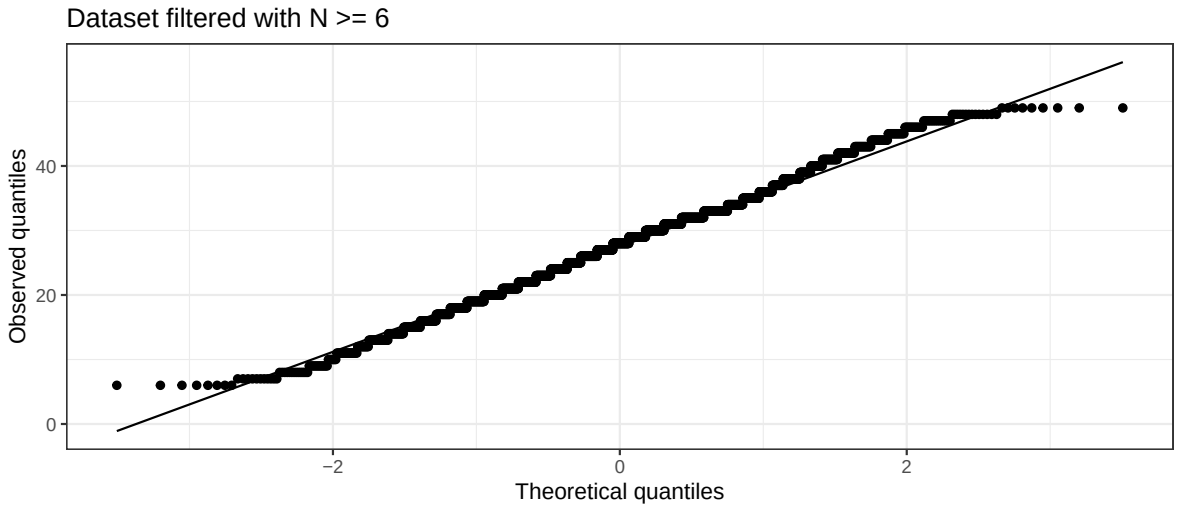
```
data250pisum_04_5 <- data250pisum_03 %>%
  group_by(ivip) %>%           # Group by genotype
  filter(n() >= 5) %>%             # Keep only groups with 5 or more repetitions
  ungroup()
data250pisum_04_5$ivip_5 <- droplevels(data250pisum_04_5$ivip)
res5 <- calculate_broad_heritability(data = data250pisum_04_5)
```

```
data250pisum_04_6 <- data250pisum_03 %>%
  group_by(ivip) %>%           # Group by genotype
  filter(n() >= 6) %>%             # Keep only groups with 6 or more repetitions
  ungroup()
data250pisum_04_6$ivip <- droplevels(data250pisum_04_6$ivip)
res5 <- calculate_broad_heritability(data = data250pisum_04_6)
```

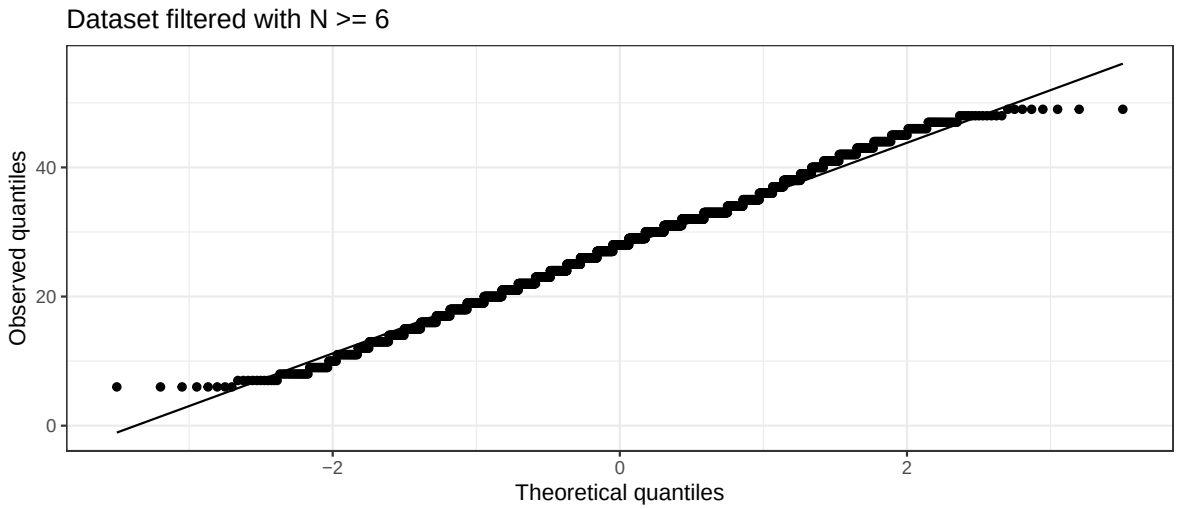
> **!** Important
>
> Conclusions:
> - Heritability is equal to 0.413.
> - There are 218 genotypes in total.
> - There are 2164 measurements in total.

Distribution normality check.

```
p <- data250pisum_04_4 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("Dataset filtered with N >= 6") +
  theme_bw()
p
```

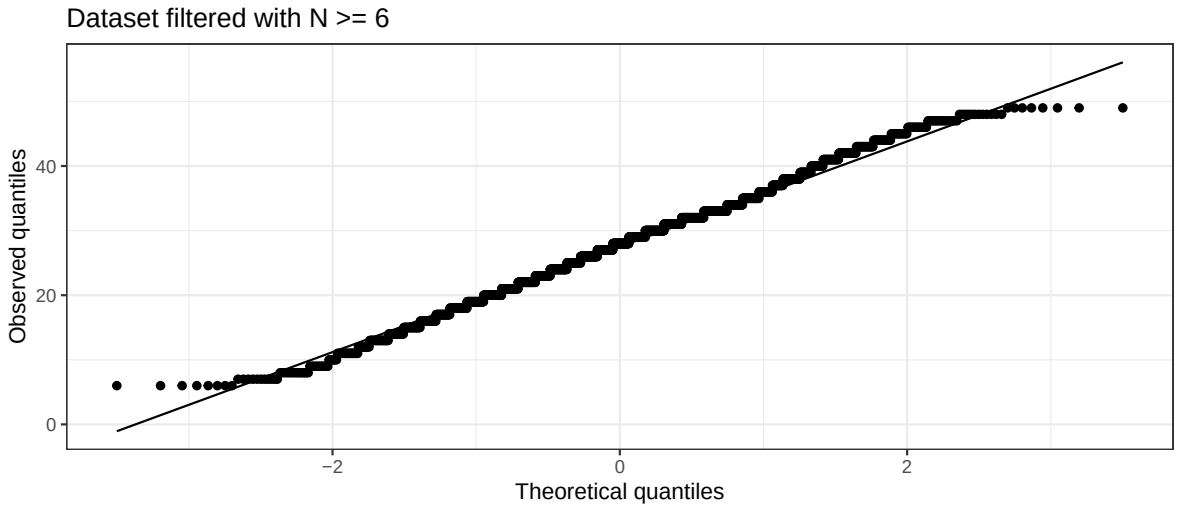



```
p <- data250pisum_04_5 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("Dataset filtered with N >= 6") +
  theme_bw()
p
```




```
p <- data250pisum_04_6 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
```

```
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("Dataset filtered with N >= 6") +
  theme_bw()
p
```

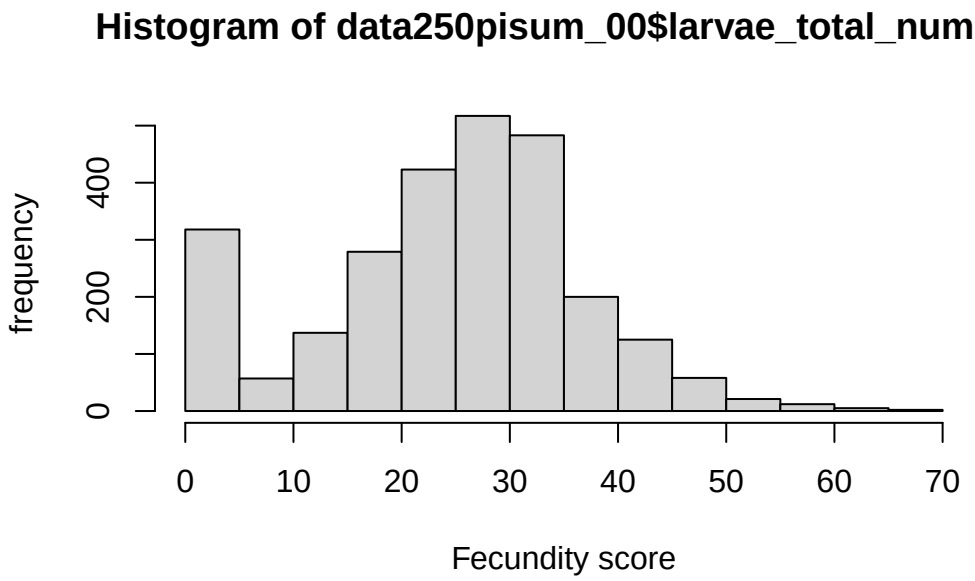



```
cor<-create_correlation_plot(data250pisum_04_6,
                         group_cols = c("genotype", "session"),
                         value_col = "larvae_total_number")
cortable<-cor$corrPos
mean(cortable$corr)
sd(cortable$corr)
mean(cortable$corr[cortable$p.value < 0.1], na.rm = TRUE)
sd(cortable$corr[cortable$p.value < 0.1], na.rm = TRUE)
```

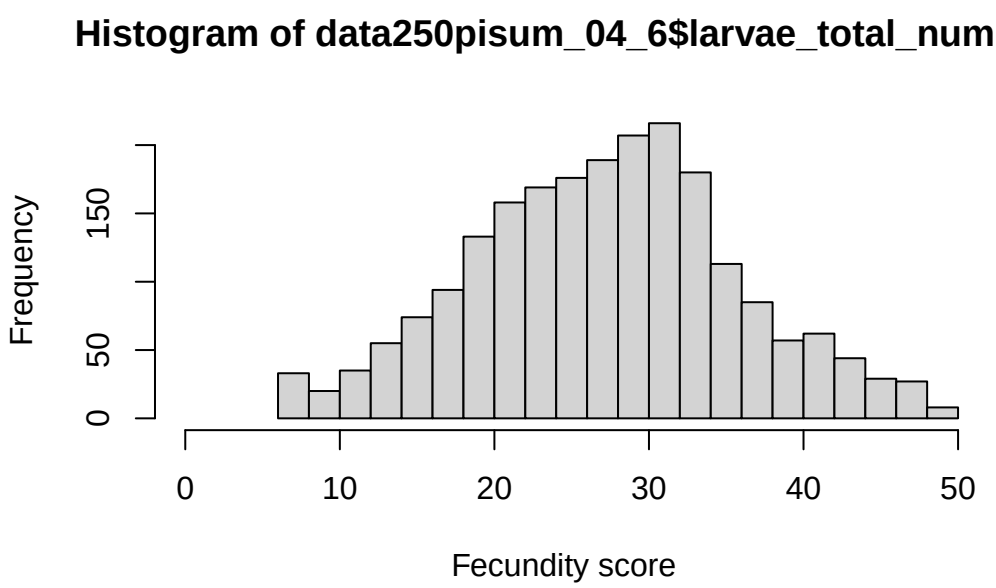
```
mod_h2 <- lmer(larvae_total_number ~ (1 | ivip) + (1 | session), data = data250pisum_04_6)
  estimates <- as.data.frame(VarCorr(mod_h2))
  genetic_variance <- estimates$vcov[1]
  residual_variance <- estimates$vcov[3]
  nbE <- nlevels(data250pisum_04_6$session)
  H2 <- genetic_variance / (genetic_variance + (residual_variance / nbE))
```

comparison between raw and final distribution

```r
hist(data250pisum_00$larvae_total_number, breaks = 20, xlab = "Fecundity score", ylab = "fre
```




```r
hist(
  data250pisum_04_6$larvae_total_number,
  breaks = 20,
  xlim = c(0, max(data250pisum_04_6$larvae_total_number, na.rm = TRUE)),
  xlab = "Fecundity score",
  ylab = "Frequency"
)
```



Write final dataset to .csv

```
write.csv(x = data250pisum_04_4,
          file = "../../../../01_phenotipic_analysis/01.dataset_filtering/02_A.pisum/final fi
          quote = F, row.names = F)
```

```
write.csv(x = data250pisum_04_5,
          file = "../../../../01_phenotipic_analysis/01.dataset_filtering/02_A.pisum/final fi
          quote = F, row.names = F)
```

```
write.csv(x = data250pisum_04_6,
          file = "../../../../01_phenotipic_analysis/01.dataset_filtering/02_A.pisum/final fi
          quote = F, row.names = F)
```